Prob 1. (8 pts) Encode 2.75 using UQ3.5 and write the result in hexadecimal.

Soln:

2.75 * 32 = 88 = 0b0101_1000 = 0x58

> -:-

Remarks of the solution:

- See Example 4 of Lctr 13.

> -:-

Prob 2. (16 pts total, 8 pts each) Decode the following Q3.4 code back to real numbers: (a) C_1 = 0x77 and (b) C_2 = 0xAA

Soln:

(a) MSB = 0 => $I_1 = C_1$, so $f_1 = I_1 \times 2^{-4}$ = 0x77 / 16 = 119/16 = 7.4375.

(b) MSB = 1 => $I_2 = C_2 - 2^8 = -86$, so $f_2 = I_2 \times 2^{-4} = -5.375$.

> -:-

Remarks of the solution:

- See Example 5 of Lctr 13.

> -:-

Prob 3. (10 pts total) Decode Q2.5 code 0b1001_1001 to real number.

Soln:

U_A = `0x99= 153`. MSB = 1 => $I_A = U_A - 256 = -103$. $f = I_A \times 2^{-5} = -3.21875$.

> -:-

Remarks of the solution:

- See Example 5 of Lctr 13.

> -:-

Prob 4. (32 pts total, 8 pts each) Encode the following real numbers into Q3.4 format and write the results in hexadecimal: (a) f_1 = 0.5, (b) f_2 = 6.73, (c) f_3 = -2.25, and (d) f_4 = -4.5.

Soln:

(a) $f_1 \times 2^4 = 0.5 \times 16 = 8$ = 0b0000_1000 = 0x08.

(b) $f_2 \times 2^4 = 6.73 \times 16 = 107.68$, $\lfloor 107.68 \rceil = 108$ = 0b0110_1100 = 0x6C.

(c) $f_3 \times 2^4 = -2.25 \times 16 = -36$ => -36 + 256 = 220 = 0b1101_1100 = 0xDC.

(d) $f_4 \times 2^4 = -4.5 \times 16 = -72$ => -72 + 256 = 184 = 0b1011_1000 = 0xB8.

> -:-

Remarks of the solution:

- See Example 7 of Lctr 13.

> -:-

Prob 5. (44 pts) Assume we use Q15 to express real numbers $f_1 = 0.5$, $f_2 = 0.25$, $f_3 = 0.125$, and $f_4 = 0.0625$. Perform the following operation in Q15: $f_A = f_1 \times f_2 + f_3 \times f_4$ where the multiplications have higher precedence than the addition. Note you need to perform the following steps:

1. (16 pts) Encode the real numbers in Q15 format to obtain I_1 to I_4.
2. (16 pts) Perform the two multiplications using 16bit x 16bit => 32bit format and shift your results as needed to obtain the two products in Q15 code I_{p1} and I_{p2}. (Like I_C in Figure 5 in Lctr 13.)
3. (6 pts) Add the two products I_{p1} and I_{p2} to obtain I_A, the Q15 code of f_A.
4. (6 pts) Decode I_A to obtain f_A.

Soln:

Part 1.

$I_1 = f_1 \times 2^{15} = 0.5 \times 2^{15} = 2^{14} = 2^2 \times 2^{12} = \text{0x4000}$.

$I_2 = f_2 \times 2^{15} = 2^{13} = \text{0x2000}$.

$I_3 = f_3 \times 2^{15} = 2^{12} = \text{0x1000}$.

$I_4 = f_4 \times 2^{15} = 2^{11} = \text{0x0800}$.

Part 2.

$I_{p1} = (I_1 \times I_2) \times 2^{-15} = (2^{14} \times 2^{13}) \times 2^{-15} = 2^{12} = \text{0x1000}$.

$I_{p2} = (I_3 \times I_4) \times 2^{-15} = (2^{12} \times 2^{11}) \times 2^{-15} = 2^{8} = \text{0x0100}$.

Part 3.

$I_A = I_{p1} + I_{p2} = \text{0x1100}$.

Part 4.

$f_A = I_A \times 2^{-15} = 2^{-3} + 2^{-7} = 0.125 + 0.0078125 = 0.1328125$.

> -:-

Remarks of the solution:

- See Example 8 of Lctr 13.

> -:-